Chapter 1

The attenuation theorems, step by step

This chapter is a line-by-line reading of Section 3 of the paper and of the proofs in its Appendix A. It states each result, then derives it: the exact attenuation of Hermite moments, the expansion of the noised log-density in the signal-to-noise ratio, the leading term of the gap between the true score and that of the covariance twin with its constant $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$, the Fisher information constants $\frac{1}{2}\|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$ and $\frac{1}{24}\|\partial_\theta \boldsymbol{\kappa}_4\|_F^2$, the general-order theorem, the sample-complexity corollary and its caveats, the crossover t_*, what happens for heavy tails, and the Edgeworth correction that puts the missing term back. The scalar case is always worked first. The notation is that of **??**: $\alpha = e^{-t}$, $\Delta_t = 1 - \alpha^2$, $\mathbf{x}_t = \alpha \mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$, and γ is the standard Gaussian on $\mathbb{R}^L$.

1.1 The statements

The paper's Section 3 contains, in order: Theorem 2 (Hermite attenuation), which is exact at every t and needs only finite moments; Assumption (E), the exponential moment; Theorem 3 (order-by-order information decay), three asymptotic equalities as $t \to \infty$ at fixed L,

$$\mathbb{E}_{P_t}\|S - S_{\mathrm{G}}\|^2 = \tfrac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}(1+\epsilon_t), \tag{1.1}$$

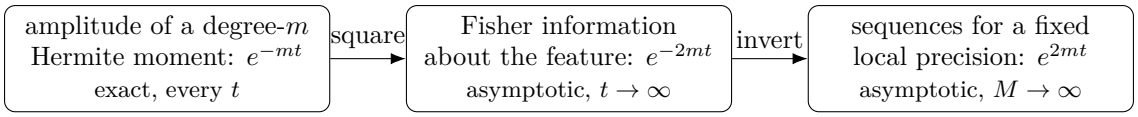
$$I_t(\theta) = \tfrac{1}{2}\|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2 e^{-4t}(1+\epsilon_t) \quad \text{if } \partial_\theta \boldsymbol{\Sigma}_0 \neq 0, \tag{1.2}$$

$$I_t(\theta) = \tfrac{1}{24}\|\partial_\theta \boldsymbol{\kappa}_4\|_F^2 e^{-8t}(1+\epsilon_t) \quad \text{if } \partial_\theta \boldsymbol{\Sigma}_0 = 0,\ \partial_\theta \boldsymbol{\kappa}_4 \neq 0, \tag{1.3}$$

with $\epsilon_t = O(e^{-2t})$; Theorem 4 (general order), which replaces the constants by the Hermite norm of the derivative of the degree-m moment tensor; Corollary 5 (sample complexity by order), a local asymptotic minimax statement; a paragraph on the crossover $t_* = \frac{1}{2}\log \lambda_{\max}(\boldsymbol{\Sigma}_0)$; and a paragraph on heavy tails, stated as an observation and not a theorem. Here S is the score of P_t, $S_{\mathrm{G}} = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$ that of the covariance twin (**??**), $\boldsymbol{\kappa}_4$ the fourth-cumulant tensor of P_0, $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ the Fisher information one noised sequence carries about a parameter θ of P_0, and $\|\cdot\|_F$ the Frobenius norm over ordered indices.

1.2 Three clocks

The paper's abstract and Section 3 use three exponentials for one feature, and the audit of 18–19 September 2026 asked that the dictionary be stated once. Figure 1.1 is that dictionary. A **feature** of P_0 is any function of its law; its **Hermite degree** m is the lowest degree of a Hermite moment $\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]$ that the feature moves. Correlation moves $\mathbb{E}[a_i a_j]$, degree two; the shape of the innovation at fixed covariance first moves the fourth cumulant, degree four. The amplitude of the moment decays as e^{-mt} exactly (Theorem 1.2); the Fisher information, being



correlation ($m = 2$): e^{-2t}	e^{-4t}	e^{4t}
shape ($m = 4$): e^{-4t}	e^{-8t}	e^{8t}

Figure 1.1: The three clocks of the paper for a feature first visible at Hermite degree m. The first is a theorem at every t; the second and third are asymptotic statements, in t and in M respectively, and the two limits do not commute (Section 1.11).

the expectation of a squared score, decays as the square, e^{-2mt} (Section 1.9); and the number of sequences needed to hold a local precision fixed is the reciprocal of the information, e^{2mt} (Section 1.11). The squared score gap Equation (1.1) is the square of a degree-four amplitude, hence e^{-8t}.

1.3 Hermite attenuation

The probabilists' **Hermite polynomials** are $\mathrm{He}_0 = 1$, $\mathrm{He}_1(x) = x$, $\mathrm{He}_2(x) = x^2 - 1$, $\mathrm{He}_3(x) = x^3 - 3x$, $\mathrm{He}_4(x) = x^4 - 6x^2 + 3$, and in general $\mathrm{He}_{n+1} = x\mathrm{He}_n - n\mathrm{He}_{n-1}$; they are generated by

$$e^{sx - s^2/2} = \sum_{n \geq 0} \mathrm{He}_n(x) \frac{s^n}{n!} \tag{1.4}$$

and are orthogonal under γ with $\mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m] = n!\, \delta_{nm}$. For a multi-index $\nu \in \mathbb{N}^L$ put $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$, of degree $|\nu| = \sum_i \nu_i$, with $\mathbb{E}_\gamma[H_\nu H_\mu] = \nu!\, \delta_{\nu\mu}$, $\nu! = \prod_i \nu_i!$. The compendium's chapter on Gaussian and Hermite structure develops these; here only Equation (1.4) and the orthogonality are used.

Lemma 1.1 (Mehler)**.** *For $z \sim \mathcal{N}(0,1)$, real a and $\alpha \in [0,1]$ with $\beta = \sqrt{1 - \alpha^2}$, $\mathbb{E}_z\big[\mathrm{He}_n(\alpha a + \beta z)\big] = \alpha^n \mathrm{He}_n(a)$ for every n.*

Proof. Put $x = \alpha a + \beta z$ in Equation (1.4) and average over z. On the left,

$$\mathbb{E}_z e^{s(\alpha a + \beta z) - s^2/2} = e^{s\alpha a - s^2/2}\, \mathbb{E}_z e^{s\beta z} = e^{s\alpha a - s^2/2 + s^2\beta^2/2} = e^{s\alpha a - s^2\alpha^2/2},$$

which by Equation (1.4) again (with $x = a$ and s replaced by αs) is $\sum_n \mathrm{He}_n(a)(\alpha s)^n/n!$. On the right, exchanging $\mathbb{E}_z$ and the sum (the series converges absolutely and uniformly on compact sets of s), $\sum_n \mathbb{E}_z[\mathrm{He}_n(\alpha a + \beta z)] s^n/n!$. Comparing coefficients of s^n gives the claim. □

Theorem 1.2 (Hermite attenuation; the paper's Theorem 2)**.** *For every multi-index ν with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$,*

$$\mathbb{E}_{P_t}\big[H_\nu(\mathbf{x}_t)\big] = e^{-|\nu| t}\, \mathbb{E}_{P_0}\big[H_\nu(\mathbf{a})\big].$$

Proof. Conditionally on $\mathbf{a}$ the coordinates $x_{t,i} = \alpha a_i + \sqrt{\Delta_t} z_i$ are independent, so $\mathbb{E}[H_\nu(\mathbf{x}_t) \mid \mathbf{a}] = \prod_i \mathbb{E}_{z_i}[\mathrm{He}_{\nu_i}(\alpha a_i + \sqrt{\Delta_t} z_i)] = \prod_i \alpha^{\nu_i} \mathrm{He}_{\nu_i}(a_i) = \alpha^{|\nu|} H_\nu(\mathbf{a})$ by Lemma 1.1 with $\beta = \sqrt{\Delta_t}$. Averaging over $\mathbf{a} \sim P_0$ needs $\mathbb{E}|H_\nu(\mathbf{a})| < \infty$, which holds because H_ν is a polynomial of degree $|\nu|$ and $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$. □

Corollary 1.3 (Consequences used by the paper)**.** *1. Cumulants of order $m \geq 3$ scale exactly:* $\kappa_m(\mathbf{x}_t) = e^{-mt} \kappa_m(\mathbf{a})$.

2. *Two laws with the same covariance $\mathbf{\Sigma}_0$ have degree-four Hermite moments that differ exactly by their fourth-cumulant tensors: for $|\nu| = 4$ with index tuple (i, j, k, l), $\mathbb{E}[H_\nu(\mathbf{a})] - \mathbb{E}'[H_\nu(\mathbf{a})] = \boldsymbol{\kappa}_4[ijkl] - \boldsymbol{\kappa}'_4[ijkl]$, attenuated by e^{-4t} under the channel.*

Proof. (i) is independent of Hermite polynomials: $\mathbf{x}_t = \alpha\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ is a sum of independent terms, joint cumulants are additive over independent summands and homogeneous of degree m under scaling, and a Gaussian has no cumulants of order three or more. (ii): H_ν is monic in the monomial $\mathbf{x}^\nu$ plus polynomials of degree at most two, so $\mathbb{E}[H_\nu(\mathbf{a})]$ is the fourth moment $\mathbb{E}[a_i a_j a_k a_l]$ plus a polynomial in second moments; the fourth moment is $\boldsymbol{\kappa}_4[ijkl] + \mathbf{\Sigma}_{ij}\mathbf{\Sigma}_{kl} + \mathbf{\Sigma}_{ik}\mathbf{\Sigma}_{jl} + \mathbf{\Sigma}_{il}\mathbf{\Sigma}_{jk}$, and everything except $\boldsymbol{\kappa}_4$ is shared by the two laws. □

Nothing here requires $P_0/\gamma \in L^2(\gamma)$, which fails for the Laplace chain and for the Gaussian chain at $\rho = 0.85$ (Section 1.12). This is the reason the paper proves Theorem 3 through the function Λ of the next section rather than through the spectral theory of the Ornstein–Uhlenbeck semigroup on $L^2(\gamma)$.

1.4 The exact identities behind the expansion

Definition 1.4 (The tilted cumulant generating function)**.** For $s \in \mathbb{R}^L$ and $b \geq 0$,

$$\Lambda(s, b) = \log \mathbb{E}_{P_0} \exp\Big(s \cdot \mathbf{a} - \frac{b}{2}|\mathbf{a}|^2\Big) \in (-\infty, +\infty]. \tag{1.5}$$

Proposition 1.5 (Exact identities; the paper's Appendix A.3, first paragraph)**.** *With $s = \alpha\mathbf{x}/\Delta_t$ and $b = \alpha^2/\Delta_t$,*

$$\log P_t(\mathbf{x}) = \log \gamma_{\Delta_t}(\mathbf{x}) + \Lambda(s, b), \tag{1.6}$$

$$\langle \mathbf{a} \rangle_{x,t} = \nabla_s \Lambda(s, b), \tag{1.7}$$

$$S(\mathbf{x}, t) = -\frac{\mathbf{x}}{\Delta_t} + \frac{\alpha}{\Delta_t}\nabla_s \Lambda(s, b), \tag{1.8}$$

$$\partial_\theta \log P_t(\mathbf{x}) = \partial_\theta \Lambda(s, b) \quad \textit{for a parameter } \theta \textit{ of } P_0. \tag{1.9}$$

These need only that P_0 be a probability measure with $\mathbb{E}|\mathbf{a}| < \infty$.

Proof. Expand the square in the Gaussian density of **??**:

$$\gamma_{\Delta_t}(\mathbf{x} - \alpha\mathbf{a}) = (2\pi\Delta_t)^{-L/2} \exp\Big(-\frac{|\mathbf{x}|^2}{2\Delta_t} + \frac{\alpha\,\mathbf{x}\cdot\mathbf{a}}{\Delta_t} - \frac{\alpha^2|\mathbf{a}|^2}{2\Delta_t}\Big) = \gamma_{\Delta_t}(\mathbf{x}) \exp\Big(s \cdot \mathbf{a} - \frac{b}{2}|\mathbf{a}|^2\Big).$$

Averaging over P_0 and taking logarithms gives Equation (1.6). The tilted law $\propto P_0(\,\mathrm{d}\mathbf{a}) \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$ is, by the same display, the posterior of $\mathbf{a}$ given $\mathbf{x}_t = \mathbf{x}$; differentiating Equation (1.5) in s under the integral gives $\nabla_s \Lambda = \mathbb{E}_{\text{tilted}}[\mathbf{a}]$, which is Equation (1.7). Then Equation (1.8) is the gradient of Equation (1.6) by the chain rule, $\nabla_{\mathbf{x}} s = (\alpha/\Delta_t)\mathbf{I}$, and it agrees with Tweedie's identity **??**. Finally γ_{Δ_t} does not depend on θ, which gives Equation (1.9). □

Example 1.6 (The Gaussian Λ)**.** For $P_0 = \mathcal{N}(0, \mathbf{\Sigma}_0)$, completing the square gives $\Lambda^{\mathrm{G}}(s, b) = -\frac{1}{2}\log\det(\mathbf{I} + b\mathbf{\Sigma}_0) + \frac{1}{2}s^\top(\mathbf{\Sigma}_0^{-1} + b\mathbf{I})^{-1}s$. In one dimension with $\mathbf{\Sigma}_0 = 1$: $\Lambda^{\mathrm{G}} = -\frac{1}{2}\log(1 + b) + s^2/(2(1 + b))$, and inserting $s = \alpha x/\Delta_t$, $b = \alpha^2/\Delta_t$ (so that $\Delta_t(1 + b) = 1$) into Equation (1.6) returns $\log P_t(x) = -\frac{1}{2}x^2 - \frac{1}{2}\log 2\pi$: the noised law of $\mathcal{N}(0, 1)$ is $\mathcal{N}(0, 1)$, as it must be.

1.5 Grading by order in e^{-t}

Both arguments of Λ are small at high noise: $s = \alpha\mathbf{x}/\Delta_t = O(\alpha)$ and $b = \alpha^2/\Delta_t = O(\alpha^2)$ for fixed $\mathbf{x}$. A monomial $s^j b^k$ is therefore of order α^{j+2k}, and the paper calls $j + 2k$ its **grade**. The expansion of Λ at the origin is a cumulant expansion. Write $X = s \cdot \mathbf{a}$ and $W = |\mathbf{a}|^2/2$, so that $\Lambda = \log \mathbb{E} e^{X - bW}$; if the expansion is legitimate (the subject of Section 1.6),

$$\Lambda = \sum_{n \geq 1} \frac{\kappa_n(X - bW)}{n!}, \qquad \kappa_n(X - bW) = \sum_{k=0}^{n} \binom{n}{k} (-b)^k \operatorname{cum}(\underbrace{X, \dots, X}_{n-k}, \underbrace{W, \dots, W}_{k}), \tag{1.10}$$

by the multilinearity of joint cumulants. The term with $n - k$ copies of X and k copies of W has grade $(n - k) + 2k = n + k$. Under the paper's standing symmetry $\mathbf{a} \mapsto -\mathbf{a}$ of P_0, a joint cumulant with an odd number of odd arguments vanishes, so $n - k$ is even and only even grades occur: $\Lambda = \Lambda_2 + \Lambda_4 + R_6$.

Proposition 1.7 (Grades two and four)**.** *With* cum *the joint cumulant under* P_0 *and* $\boldsymbol{\Sigma}_0 = \operatorname{Cov}(\mathbf{a})$*,*

$$\Lambda_2 = \tfrac{1}{2} s^\top \boldsymbol{\Sigma}_0 s - \tfrac{b}{2} \operatorname{tr} \boldsymbol{\Sigma}_0, \tag{1.11}$$

$$\Lambda_4 = \tfrac{1}{24} \boldsymbol{\kappa}_4[s, s, s, s] - \tfrac{b}{4} \sum_i \operatorname{cum}(s \cdot \mathbf{a},\, s \cdot \mathbf{a},\, a_i^2) + \tfrac{b^2}{8} \sum_{ij} \operatorname{cum}(a_i^2,\, a_j^2), \tag{1.12}$$

where $\boldsymbol{\kappa}_4[s, s, s, s] = \sum_{ijkl} \boldsymbol{\kappa}_4[ijkl] s_i s_j s_k s_l$*. Moreover*

$$\operatorname{cum}(s \cdot \mathbf{a}, s \cdot \mathbf{a}, a_i^2) = \boldsymbol{\kappa}_4[s, s, i, i] + 2\,(\boldsymbol{\Sigma}_0 s)_i^2, \qquad \operatorname{cum}(a_i^2, a_j^2) = \boldsymbol{\kappa}_4[iijj] + 2\,\boldsymbol{\Sigma}_{0,ij}^2, \tag{1.13}$$

so that Λ_2 *is the same for* P_0 *and its covariance twin, and*

$$\Lambda_4 - \Lambda_4^{\mathrm{G}} = \tfrac{1}{24} \boldsymbol{\kappa}_4[s, s, s, s] - \tfrac{b}{4} \sum_i \boldsymbol{\kappa}_4[s, s, i, i] + \tfrac{b^2}{8} \sum_{ij} \boldsymbol{\kappa}_4[iijj] \tag{1.14}$$

is linear in $\boldsymbol{\kappa}_4$ *and vanishes if and only if* $\boldsymbol{\kappa}_4 = 0$*.*

Proof. Grade two collects $(n, k) = (2, 0)$, giving $\kappa_2(X)/2 = \frac{1}{2} \operatorname{Var}(s \cdot \mathbf{a}) = \frac{1}{2} s^\top \boldsymbol{\Sigma}_0 s$, and $(n, k) = (1, 1)$, giving $-b\,\mathbb{E}W = -\frac{b}{2} \operatorname{tr} \boldsymbol{\Sigma}_0$. Grade four collects $(4, 0)$: $\kappa_4(X)/24 = \boldsymbol{\kappa}_4[s, s, s, s]/24$, since the fourth cumulant of a linear form is the contraction of the cumulant tensor; $(3, 1)$ and $(2, 2)$:

$$\binom{3}{1}(-b)\frac{\operatorname{cum}(X, X, W)}{3!} = -\frac{b}{2} \operatorname{cum}(X, X, W) = -\frac{b}{4} \sum_i \operatorname{cum}(X, X, a_i^2),$$

$$\binom{2}{2} b^2 \frac{\operatorname{cum}(W, W)}{2!} = \frac{b^2}{2} \operatorname{Var}(W) = \frac{b^2}{8} \sum_{ij} \operatorname{cum}(a_i^2, a_j^2).$$

For Equation (1.13), use the formula for a cumulant with a product argument (Leonov and Shiryaev): $\operatorname{cum}(X, X, a_i^2)$ is the sum over partitions of the four variables X, X, a_i, a_i that are indecomposable with respect to the grouping $\{X\}, \{X\}, \{a_i, a_i\}$ of products of cumulants over blocks. The full block gives $\operatorname{cum}(X, X, a_i, a_i) = \boldsymbol{\kappa}_4[s, s, i, i]$; the two pairings that join each X to one a_i give $2 \operatorname{cum}(X, a_i)^2 = 2(\boldsymbol{\Sigma}_0 s)_i^2$; the pairing $\{X, X\}, \{a_i, a_i\}$ is decomposable and excluded; means vanish. Likewise

$$\operatorname{cum}(a_i^2, a_j^2) = \operatorname{Cov}(a_i^2, a_j^2) = \mathbb{E}[a_i^2 a_j^2] - \boldsymbol{\Sigma}_{ii} \boldsymbol{\Sigma}_{jj} = \boldsymbol{\kappa}_4[iijj] + 2\boldsymbol{\Sigma}_{ij}^2.$$

The twin has the same $\boldsymbol{\Sigma}_0$ and $\boldsymbol{\kappa}_4 = 0$, so subtracting gives Equation (1.14). It is a nonzero linear functional of $\boldsymbol{\kappa}_4$ unless $\boldsymbol{\kappa}_4 = 0$, because its first term alone is a nonzero quartic form in s whenever the symmetric tensor is nonzero. □

Remark 1.8 (On the paper's displayed Λ_4). The paper's Appendix A.3 prints the second and third terms of Equation (1.12) with the fourth-order joint cumulants $\operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i, a_i)$ and $\operatorname{cum}(a_i, a_i, a_j, a_j)$ in place of the cumulants of the squares. By Equation (1.13) the two differ by $-\frac{b}{2}|\boldsymbol{\Sigma}_0 s|^2 + \frac{b^2}{4}\|\boldsymbol{\Sigma}_0\|_F^2$, which is the grade-four part of Λ^{G}: the scalar Gaussian of Example 1.6 has $\Lambda_4^{\mathrm{G}} = \frac{b^2}{4} - \frac{b}{2}s^2 \neq 0$, whereas the printed formula gives zero for any Gaussian. The omission is common to P_0 and its twin, so Equation (1.14) and everything that follows from it are unaffected; the displayed Λ_4 itself is not correct as printed.

1.5.1 From the grade-four difference to a Hermite polynomial

Substituting $s = \alpha\mathbf{x}/\Delta_t = \alpha\mathbf{x}(1+\alpha^2+\dots)$ and $b = \alpha^2/\Delta_t = \alpha^2(1+\alpha^2+\dots)$ into Equation (1.14) and keeping the leading order,

$$\Lambda_4 - \Lambda_4^{\mathrm{G}} = \alpha^4\Big[\tfrac{1}{24}\boldsymbol{\kappa}_4[\mathbf{x},\mathbf{x},\mathbf{x},\mathbf{x}] - \tfrac{1}{4}\sum_i \boldsymbol{\kappa}_4[\mathbf{x},\mathbf{x},i,i] + \tfrac{1}{8}\sum_{ij}\boldsymbol{\kappa}_4[iijj]\Big] + O(\alpha^6(1+|\mathbf{x}|)^4).$$

The bracket is a Hermite form. The **Wick product** (multivariate Hermite polynomial) of four coordinates of the standard Gaussian is

$$\begin{aligned} :x_ix_jx_kx_l: = {} & x_ix_jx_kx_l - \big(\delta_{ij}x_kx_l + \delta_{ik}x_jx_l + \delta_{il}x_jx_k + \delta_{jk}x_ix_l + \delta_{jl}x_ix_k + \delta_{kl}x_ix_j\big) \\ & + \big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big), \end{aligned} \tag{1.15}$$

which reduces to H_ν when the indices are grouped into a multi-index (for instance $:x_1^4: = \mathrm{He}_4(x_1)$ and $:x_1^2x_2^2: = \mathrm{He}_2(x_1)\mathrm{He}_2(x_2)$). Contracting Equation (1.15) with a symmetric tensor, the six single contractions are equal and the three double contractions are equal, so $\frac{1}{24}\sum_{ijkl}\boldsymbol{\kappa}_4[ijkl]:x_ix_jx_kx_l:$ is exactly the bracket. Hence

$$\Lambda_4 - \Lambda_4^{\mathrm{G}} = \frac{\alpha^4}{24}\sum_{ijkl}\boldsymbol{\kappa}_4[ijkl]\,:x_ix_jx_kx_l: + O\big(\alpha^6(1+|\mathbf{x}|)^4\big), \tag{1.16}$$

which is the paper's "the three contributions of Λ_4 combine into the Hermite form". In one dimension with unit variance the bracket is $\kappa_4[\frac{1}{24}x^4 - \frac{1}{4}x^2 + \frac{1}{8}] = \frac{\kappa_4}{24}(x^4 - 6x^2 + 3) = \frac{\kappa_4}{24}\mathrm{He}_4(x)$.

1.6 Where the expansion lives: the exponential moment

Assumption 1.9 (Exponential moment; the paper's Assumption (E))**.** $\mathbb{E}_{P_0}e^{\lambda|\mathbf{a}|} < \infty$ for some $\lambda > 0$.

Gaussian, Laplace, uniform, bounded mixtures and the generalised Gaussian with $\beta \geq 1$ satisfy it; Student-t does not.

Lemma 1.10 (Analyticity and uniform tilted moments)**.** *Under Assumption 1.9: (i) $\Lambda(s,b)$ is finite on $\{|s| \leq \lambda\} \times \{b \geq 0\}$ and real-analytic in (s,b) on $\{|s| < \lambda\} \times \{b \geq 0\}$, with Taylor coefficients at the origin equal to the joint cumulants of $X = s\cdot\mathbf{a}$ and $W = |\mathbf{a}|^2/2$, so that Equation (1.10) is a convergent expansion there; (ii) the tilted laws $\propto P_0(\,\mathrm{d}\mathbf{a})\,e^{\vartheta(s\cdot\mathbf{a} - b|\mathbf{a}|^2/2)}$, $\vartheta \in [0,1]$, have moments of every order bounded uniformly on $\{|s| \leq \lambda/2,\ 0 \leq b \leq 1\}$ by constants that depend only on λ and $\mathbb{E}e^{\lambda|\mathbf{a}|}$.*

Proof. (i) For $|s| \leq \lambda$ and $b \geq 0$, $|e^{s\cdot\mathbf{a} - b|\mathbf{a}|^2/2}| \leq e^{\lambda|\mathbf{a}|}$, integrable. For complex s with $|\operatorname{Re} s| < \lambda$ the same bound holds, the integrand is entire in s and in b, and dominated convergence makes the integral holomorphic; a holomorphic function of the complex variables that is real on the real set is real-analytic there, and the logarithm of a positive real-analytic function is real-analytic. The Taylor coefficients of $\log\mathbb{E}e^{X-bW}$ are the joint cumulants by the definition of cumulants as the coefficients of the log-moment generating function. (ii) For $|s| \leq \lambda/2$ and $0 \leq b \leq 1$ the tilting weight is at most $e^{\lambda|\mathbf{a}|/2}$, so $\mathbb{E}_{\mathrm{tilted}}|\mathbf{a}|^k \leq \mathbb{E}[|\mathbf{a}|^k e^{\lambda|\mathbf{a}|/2}]/Z$ with $|\mathbf{a}|^k e^{\lambda|\mathbf{a}|/2} \leq C_k(\lambda)e^{\lambda|\mathbf{a}|}$; the normaliser $Z = \mathbb{E}e^{\vartheta(X-bW)}$ is at least $\exp(\mathbb{E}[\vartheta(X-bW)]) \geq \exp(-\frac{1}{2}\mathbb{E}|\mathbf{a}|^2)$ by Jensen, and $\mathbb{E}|\mathbf{a}|^2$ is bounded by $\mathbb{E}e^{\lambda|\mathbf{a}|}$ and λ. □

Why finite moments of all orders do not suffice. The audit of 18–19 September 2026 (`docs/SCIENCE_AUDIT.md`; `reports/audit-2026-09-18.tex`, Section 3.3) found that an earlier draft stated Theorem 3 under "finite eighth moments" while its proof used analyticity and an exponential tail bound; the theorem is now stated under Assumption 1.9. Three things break without it, and the third is not repairable by adding more finite moments.

1. The expansion is taken at $b = 0$, where $\Lambda(s, 0) = \log \mathbb{E}e^{s\cdot\mathbf{a}}$ is the log-moment generating function. For a law with polynomial tails, such as Student-t, it is $+\infty$ for every $s \neq 0$: the Taylor coefficients (the cumulants) exist up to the order of the finite moments, but there is no function for them to expand. Even a law with *all* moments finite can have $\mathbb{E}e^{\lambda|\mathbf{a}|} = \infty$ for every $\lambda > 0$ (a density with tails $\propto e^{-|a|^{1/2}}$ is an example); its cumulants then grow so fast that the Taylor series at the origin has radius of convergence zero, and $\Lambda(\cdot, 0)$ is smooth but not analytic.

2. The Lagrange remainder R_6 is a sixth derivative of Λ at an intermediate point $(\vartheta s, \vartheta b)$, i.e. a tilted sixth-order joint cumulant. On the bulk $\{|s| \leq \lambda/2\}$ these are uniformly bounded by Lemma 1.10(ii); without Assumption 1.9 there is no bulk of fixed size on which they are, and for $\vartheta \to 0$ they tend to the untilted sixth cumulant, which for Student-t_5 is infinite.

3. The tail of the bulk must have probability below every power of α for the remainder to be negligible at the order α^8 of the theorem. Under Assumption 1.9 it is $\exp(-c\,e^{2t})$ (Section 1.8); under finite moments of order p it is only $O(\alpha^{2p})$ by Markov's inequality, polynomial in α, and for a regularly varying tail of index ν the tail's own contribution to the gap is of order α^ν, which leads whenever $\nu < 8$ (Section 1.13).

The audit also observes that the *conclusion* Equation (1.1) may well hold under a moment condition of order strictly greater than eight (the heuristic of Section 1.13 makes finite eighth moments the borderline), but the present proof does not show it, and the paper claims the theorem only under Assumption 1.9.

1.7 The leading term of the score gap and Wick's theorem

1.7.1 The scalar computation in full

Let $L = 1$, $\mathrm{Var}(a) = 1$ and κ_4 the fourth cumulant of P_0. By Equation (1.8) and Equation (1.16),

$$\begin{aligned} S - S_{\mathrm{G}} &= \frac{\alpha}{\Delta_t}\,\partial_s(\Lambda - \Lambda^{\mathrm{G}}) = \partial_x(\Lambda - \Lambda^{\mathrm{G}}) \\ &= \partial_x\Big[\frac{\kappa_4\alpha^4}{24}\mathrm{He}_4(x)\Big] + O(\alpha^6(1+|x|)^5) = \frac{\kappa_4}{6}\mathrm{He}_3(x)\,\alpha^4 + O(\alpha^6(1+|x|)^5), \end{aligned}$$

using $\partial_x = (\alpha/\Delta_t)\partial_s$ and $\mathrm{He}_4' = 4\mathrm{He}_3$. This is the paper's "the scalar leading term of (3) is $(\kappa_4/6)\mathrm{He}_3(x)e^{-4t}$". Its γ-second moment is

$$\mathbb{E}_\gamma\Big[\Big(\frac{\kappa_4}{6}\mathrm{He}_3\Big)^2\Big]\alpha^8 = \frac{\kappa_4^2}{36}\,3!\,\alpha^8 = \frac{\kappa_4^2}{6}\alpha^8,$$

which is Equation (1.1) with $\|\kappa_4\|_F^2 = \kappa_4^2$.

1.7.2 Wick's theorem

Theorem 1.11 (Wick)**.** *Under γ, for index tuples $J = (j_1, \ldots, j_p)$ and $K = (k_1, \ldots, k_q)$,*

$$\mathbb{E}_\gamma[{:}x_{j_1}\cdots x_{j_p}{:}\,{:}x_{k_1}\cdots x_{k_q}{:}] = \delta_{pq}\sum_{\sigma\in S_p}\prod_{r=1}^{p}\delta_{j_r,k_{\sigma(r)}},$$

the number of perfect matchings between the two tuples that pair equal indices.

This is the orthogonality $\mathbb{E}_\gamma[H_\nu H_\mu] = \nu!\delta_{\nu\mu}$ written for ordered tuples: when both tuples are the same multiset ν, the matchings that pair equal indices number $\nu!$.

Proposition 1.12 (The constant $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$)**.** *The m-th component of the leading term of $S - S_{\mathrm{G}}$ is $\alpha^4 \frac{1}{6}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]\,{:}x_j x_k x_l{:}$, and*

$$\mathbb{E}_\gamma\Big[\Big(\tfrac{1}{6}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]{:}x_j x_k x_l{:}\Big)^2\Big] = \frac{3!}{36}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]^2, \qquad \sum_m(\cdots) = \tfrac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2.$$

Proof. Differentiating Equation (1.16) with respect to x_m: each of the four index slots can carry m, and $\partial_{x_m}{:}x_i x_j x_k x_l{:} = \delta_{im}{:}x_j x_k x_l{:} + \dots$ (the derivative of a Wick monomial removes one factor, as for Hermite polynomials, $\mathrm{He}_n' = n\mathrm{He}_{n-1}$); by symmetry of $\boldsymbol{\kappa}_4$ the four contributions are equal, giving $\frac{4}{24} = \frac{1}{6}$. For the second moment, expand the square and apply Theorem 1.11 with $p = q = 3$:

$$\begin{aligned}\sum_{jkl}\sum_{j'k'l'} \boldsymbol{\kappa}_4[mjkl]\boldsymbol{\kappa}_4[mj'k'l'] \sum_{\sigma\in S_3}\prod_r \delta_{(jkl)_r,(j'k'l')_{\sigma(r)}} &= \sum_{\sigma\in S_3}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]\boldsymbol{\kappa}_4[m\,\sigma(jkl)] \\ &= 3!\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]^2,\end{aligned}$$

since $\boldsymbol{\kappa}_4$ is symmetric under permutation of its slots. Multiply by $\frac{1}{36}$ and sum over m; the sum over all four ordered indices of $\boldsymbol{\kappa}_4[mjkl]^2$ is $\|\boldsymbol{\kappa}_4\|_F^2$. □

1.7.3 From γ to P_t, and the remainder

The theorem is a statement under P_t, not γ. Two corrections are needed on the bulk, both of relative size $O(\alpha^2)$, and both are what the factor $(1 + \epsilon_t)$ absorbs.

- *Change of measure.* By Mehler's kernel (Section 1.10), $P_t/\gamma = 1 + \frac{\alpha^2}{2}\langle \boldsymbol{\Sigma}_0 - \mathbf{I}, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle + O(\alpha^4(1 + |\mathbf{x}|)^4)$. Integrating the square of the degree-three leading term against the degree-two correction gives a term of relative order α^2. The paper says the correction "has zero γ-mean against the square of a degree-3 polynomial up to terms of relative size α^2"; in fact its γ-mean is not zero in general (in one dimension $\mathrm{He}_3^2 = \mathrm{He}_6 + 9\mathrm{He}_4 + 18\mathrm{He}_2 + 6$, so $\mathbb{E}_\gamma[\mathrm{He}_2\mathrm{He}_3^2] = 36$), but it carries the explicit factor α^2, which is all the theorem needs.
- *Remainder.* The cross term between the leading term (order α^4) and $\nabla_s R_6$ (order α^6 times a polynomial in $|\mathbf{x}|$, by Lemma 1.10(ii)) is $O(\alpha^{10})$ by Cauchy–Schwarz, and the square of the remainder is $O(\alpha^{12})$.

The constant in $O(\alpha^2)$ depends on P_0 and on L; for the Gaussian chain it is $\lambda_{\max}(\boldsymbol{\Sigma}_0)$ (Section 1.12).

1.8 The tail

The expansion is used only on the **bulk** $B = \{|s| \le \lambda/2\} = \{|\mathbf{x}| \le \lambda\Delta_t/(2\alpha)\}$, a ball whose radius grows like e^t.

Lemma 1.13 (Tail probability)**.** *Under Assumption 1.9, with $c = \lambda\Delta_t/4$,*

$$P_t(B^c) \le e^{-\lambda c/\alpha^2}\,\mathbb{E}e^{\lambda|\mathbf{a}|} + \mathbb{P}(\sqrt{\Delta_t}|\mathbf{z}| > c/\alpha),$$

and the second term is at most $C_L e^{-c^2/(4\alpha^2)}$; both are $\exp(-c'e^{2t})$.

Proof. On B^c, $|\alpha\mathbf{x}| > \lambda\Delta_t/2 = 2c$. Since $\alpha\mathbf{x} = \alpha^2\mathbf{a} + \alpha\sqrt{\Delta_t}\mathbf{z}$, either $\alpha^2|\mathbf{a}| > c$ or $\alpha\sqrt{\Delta_t}|\mathbf{z}| > c$. For the first, $\mathbb{P}(|\mathbf{a}| > c/\alpha^2) \le e^{-\lambda c/\alpha^2}\mathbb{E}e^{\lambda|\mathbf{a}|}$ by Markov's inequality applied to $e^{\lambda|\mathbf{a}|}$. For the second, $|\mathbf{z}|^2$ is χ^2 with L degrees of freedom and $\mathbb{E}e^{|\mathbf{z}|^2/4} = 2^{L/2}$, so $\mathbb{P}(|\mathbf{z}| > R) \le 2^{L/2}e^{-R^2/4}$ with $R = c/(\alpha\sqrt{\Delta_t})$. □

Proposition 1.14 (The tail contributes below every power of α)**.** $\mathbb{E}_{P_t}[\|S - S_{\mathrm{G}}\|^2 \mathbf{1}_{B^c}] = O(e^{-c'' e^{2t}})$ *for some* $c'' > 0$.

Proof. By **??** and **??**, $\|S - S_{\mathrm{G}}\|^2 \le C(|\mathbf{x}|^2 + \alpha^2 \|\langle \mathbf{a} \rangle_{x,t}\|^2 + \alpha^2 \|\alpha \boldsymbol{\Sigma}_0 \boldsymbol{\Sigma}_t^{-1} \mathbf{x}\|^2)/\Delta_t^2$. Cauchy–Schwarz gives $\mathbb{E}[\|\cdot\|^2 \mathbf{1}_{B^c}] \le (\mathbb{E}\|\cdot\|^4)^{1/2} P_t(B^c)^{1/2}$, and the fourth moment is finite: $\mathbb{E}|\mathbf{x}|^4 < \infty$, and $\mathbb{E}\|\langle \mathbf{a} \rangle_{x,t}\|^4 \le \mathbb{E}|\mathbf{a}|^4$ by Jensen's inequality for the conditional expectation. The tail probability is $\exp(-c' e^{2t})$ by Lemma 1.13, and its square root still is. □

Putting Proposition 1.12, the bulk corrections and Proposition 1.14 together gives Equation (1.1): the paper's Theorem 3, first line. The paper notes what changes without Assumption 1.9: the tail probability becomes polynomial in α, of order $\alpha^{2\nu}$ for a tail index ν, and the tail's own contribution is of order α^{ν}; this is Section 1.13.

1.9 Fisher information: degree two and degree four

By Equation (1.9), $\partial_\theta \log P_t = \partial_\theta \Lambda$, and the graded expansion of Λ gives the graded expansion of the score with respect to θ.

Proposition 1.15 (Degree-two constant)**.** *If* $\partial_\theta \boldsymbol{\Sigma}_0 \neq 0$*, the leading term of* $\partial_\theta \log P_t$ *is*

$$\tfrac{\alpha^2}{2} \langle \partial_\theta \boldsymbol{\Sigma}_0, :\mathbf{x}\mathbf{x}^\top: \rangle = \tfrac{\alpha^2}{2} \sum_{ij} \partial_\theta \boldsymbol{\Sigma}_{0,ij} \, :x_i x_j:,$$

and its γ*-second moment is* $\frac{\alpha^4}{2} \|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$.

Proof. From Equation (1.11) with $s \approx \alpha \mathbf{x}$, $b \approx \alpha^2$,

$$\partial_\theta \Lambda_2 = \tfrac{\alpha^2}{2} (\mathbf{x}^\top \partial_\theta \boldsymbol{\Sigma}_0 \mathbf{x} - \operatorname{tr} \partial_\theta \boldsymbol{\Sigma}_0) = \tfrac{\alpha^2}{2} \sum_{ij} \partial_\theta \boldsymbol{\Sigma}_{0,ij} \, (x_i x_j - \delta_{ij}),$$

and $:x_i x_j: = x_i x_j - \delta_{ij}$. By Theorem 1.11 with $p = q = 2$, $\mathbb{E}_\gamma[:x_i x_j: :x_{i'} x_{j'}:] = \delta_{ii'}\delta_{jj'} + \delta_{ij'}\delta_{ji'}$, so the second moment is

$$\tfrac{\alpha^4}{4} \sum_{ij} (\partial \boldsymbol{\Sigma}_{ij}^2 + \partial \boldsymbol{\Sigma}_{ij} \partial \boldsymbol{\Sigma}_{ji}) = \tfrac{\alpha^4}{4} \cdot 2 \|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$$

by symmetry. □

Proposition 1.16 (Degree-four constant)**.** *If* $\partial_\theta \boldsymbol{\Sigma}_0 = 0$ *and* $\partial_\theta \boldsymbol{\kappa}_4 \neq 0$*, the leading term is* $\frac{\alpha^4}{24} \sum_{ijkl} \partial_\theta \boldsymbol{\kappa}_4[ijkl] :x_i x_j x_k x_l:$*, with* γ*-second moment* $\frac{4!}{24^2} \|\partial_\theta \boldsymbol{\kappa}_4\|_F^2 \alpha^8 = \frac{1}{24} \|\partial_\theta \boldsymbol{\kappa}_4\|_F^2 \alpha^8$.

Proof. The grade-two term has vanishing θ-derivative. Differentiating Equation (1.16) in θ (only $\boldsymbol{\kappa}_4$ depends on θ) gives the leading term; Theorem 1.11 with $p = q = 4$ and the symmetry of $\partial_\theta \boldsymbol{\kappa}_4$ give $4! \, \|\partial_\theta \boldsymbol{\kappa}_4\|_F^2$ for the sum over matchings, exactly as in Proposition 1.12. □

The bulk corrections and the tail are handled as in Sections 1.7 and 1.8, which gives Equations (1.2) and (1.3).

Proposition 1.17 (The Gaussian chain, exactly)**.** *If* $P_0 = \mathcal{N}(0, \boldsymbol{\Sigma}_0(\theta))$ *then* $\mathbf{x}_t \sim \mathcal{N}(0, \boldsymbol{\Sigma}_t(\theta))$ *with* $\boldsymbol{\Sigma}_t = \alpha^2 \boldsymbol{\Sigma}_0 + \Delta_t \mathbf{I}$*, and at every* t

$$I_t(\theta) = \tfrac{1}{2} \operatorname{tr}(\boldsymbol{\Sigma}_t^{-1} \partial_\theta \boldsymbol{\Sigma}_t \, \boldsymbol{\Sigma}_t^{-1} \partial_\theta \boldsymbol{\Sigma}_t), \qquad \partial_\theta \boldsymbol{\Sigma}_t = \alpha^2 \partial_\theta \boldsymbol{\Sigma}_0,$$

so that $e^{4t} I_t(\theta) \to \frac{1}{2} \|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$ *with relative correction* $O(\alpha^2 \lambda_{\max}(\boldsymbol{\Sigma}_0))$.

Proof. The Fisher information of a centred Gaussian family about a parameter of its covariance is $\frac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial\boldsymbol{\Sigma}\,\boldsymbol{\Sigma}^{-1}\partial\boldsymbol{\Sigma})$ (differentiate $\log\det\boldsymbol{\Sigma}$ and $\mathbf{x}^\top\boldsymbol{\Sigma}^{-1}\mathbf{x}$ twice and take expectations). With $\boldsymbol{\Sigma}_t = \mathbf{I} + \alpha^2(\boldsymbol{\Sigma}_0 - \mathbf{I})$ the Neumann series gives $\boldsymbol{\Sigma}_t^{-1} = \mathbf{I} - \alpha^2(\boldsymbol{\Sigma}_0 - \mathbf{I}) + O(\alpha^4)$ in operator norm once $\alpha^2\|\boldsymbol{\Sigma}_0 - \mathbf{I}\| < 1$, hence $\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\partial\boldsymbol{\Sigma}_0) = \|\partial\boldsymbol{\Sigma}_0\|_F^2(1 + O(\alpha^2\|\boldsymbol{\Sigma}_0 - \mathbf{I}\|))$, and $\|\boldsymbol{\Sigma}_0 - \mathbf{I}\| \le \lambda_{\max}(\boldsymbol{\Sigma}_0)$. □

The code computes this closed form (`gaussian_chain_fisher` in `src/markov_score/theory.py`), and the paper's Figure (theory), panel (a), plots $I_t e^{4t}$ divided by the constant for five correlations: every curve reaches 1, and how fast is governed by t_*.

Efficient information and nuisance parameters. When θ is estimated together with other coordinates η, the relevant quantity is the **efficient information** $I_t^{\mathrm{eff}}(\theta) = I_{\theta\theta} - I_{\theta\eta}I_{\eta\eta}^{-1}I_{\eta\theta}$, the reciprocal of the θ-entry of the inverse information matrix. For a degree-four θ and degree-two nuisances η the cross information $I_{\theta\eta} = \mathbb{E}[\partial_\theta \log P_t\, \partial_\eta \log P_t]$ has no α^6 term, because the leading terms are a degree-four and a degree-two Hermite polynomial, which are γ-orthogonal (Theorem 1.11 with $p \neq q$); so $I_{\theta\eta} = O(\alpha^8)$, $I_{\eta\eta} = \Theta(\alpha^4)$, and the correction is $O(\alpha^{12})$, negligible against $I_{\theta\theta} = \Theta(\alpha^8)$. This is the paper's "the leading orders of a degree-2 and a degree-4 coordinate do not couple". A nuisance of the *same* degree (another shape coordinate, as in the eight-component mixture the paper fits) has a cross term of the same order and reduces the efficient information by a factor of order one; that is why the paper's Section 4 calls its one-parameter Cramér–Rao reference a yardstick and not an efficiency bound. The code's `efficient_information` in `src/markov_score/information.py` inverts the outer-product estimate of the full matrix.

1.10 The general order

Theorem 1.18 (General order; the paper's Theorem 4)**.** *Let P_0^θ satisfy Assumption 1.9 uniformly near θ_0, and let m be the smallest degree at which $\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})] \neq 0$ for some $|\nu| = m$. Then, at fixed L,*

$$I_t(\theta) = e^{-2mt} \sum_{|\nu|=m} \frac{\left(\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})]\right)^2}{\nu!}\,(1 + O(e^{-2t})).$$

The proof rests on a second expansion of the same function, in Hermite polynomials rather than in cumulants.

Lemma 1.19 (Mehler's kernel)**.** *Formally, and in $L^2(\gamma)$ whenever $\sum_\nu \alpha^{2|\nu|}(\mathbb{E}_{P_0}H_\nu)^2/\nu! < \infty$,*

$$\frac{P_t(\mathbf{x})}{\gamma(\mathbf{x})} = \sum_\nu \alpha^{|\nu|}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]\,\frac{H_\nu(\mathbf{x})}{\nu!} = 1 + \frac{\alpha^2}{2}\langle \boldsymbol{\Sigma}_0 - \mathbf{I}, :\mathbf{x}\mathbf{x}^\top:\rangle + \dots \tag{1.17}$$

Proof. In one dimension the transition density of the channel divided by γ is $\gamma_{\Delta_t}(x - \alpha a)/\gamma(x) = (1-\alpha^2)^{-1/2}\exp(-\frac{(x-\alpha a)^2}{2(1-\alpha^2)} + \frac{x^2}{2}) = \sum_n \alpha^n \mathrm{He}_n(a)\mathrm{He}_n(x)/n!$, Mehler's formula, which follows from Equation (1.4) by integrating $e^{sa - s^2/2}e^{ux - u^2/2}$ against the joint Gaussian of (a, x) with correlation α. Multiply over coordinates, average over $\mathbf{a} \sim P_0$, and read off the $|\nu| \le 2$ terms: $\mathbb{E}[a_i] = 0$, $\mathbb{E}[\mathrm{He}_2(a_i)] = \boldsymbol{\Sigma}_{0,ii} - 1$, $\mathbb{E}[a_i a_j] = \boldsymbol{\Sigma}_{0,ij}$. □

Proof of Theorem 1.18. Under Assumption 1.9 the graded expansion of Λ is a convergent series in α on the bulk, and $\log(P_t/\gamma_{\Delta_t}) = \Lambda$. On the other hand

$$\log(P_t/\gamma) = \log\Big(1 + \sum_{|\nu|\ge 2} \alpha^{|\nu|}\,\mathbb{E}[H_\nu]\,\frac{H_\nu(\mathbf{x})}{\nu!}\Big)$$

is a second formal series in α for the same function up to the explicit $\log(\gamma_{\Delta_t}/\gamma)$. At each grade both coefficients are polynomials in the moments of P_0 up to that grade; on compactly supported P_0 both series converge and represent the function, so the two polynomials agree on the class of moment sequences of compactly supported laws, which determines a polynomial identity; hence they agree identically. Now differentiate in θ. The grade-m coefficient of $\partial_\theta \log(P_t/\gamma)$ is

$$\alpha^m \sum_{|\nu|=m} \partial_\theta \mathbb{E}[H_\nu] \, \frac{H_\nu(\mathbf{x})}{\nu!}$$

plus ∂_θ of products of lower-grade terms of the logarithm's series; every such product contains a factor $\mathbb{E}[H_\mu]$ with $|\mu| < m$, whose θ-derivative vanishes by the minimality of m, and all terms of grade below m vanish for the same reason. The γ-second moment of the leading term is $\alpha^{2m} \sum_{|\nu|=m} (\partial_\theta \mathbb{E} H_\nu)^2/\nu!$ by orthogonality, and the passage from γ to P_t on the bulk and the tail are as in Sections 1.7 and 1.8. □

Consistency with the two constants. For $m = 2$: the multi-indices are $\nu = 2\mathbf{e}_i$, with $H_\nu = \mathrm{He}_2(x_i)$, $\mathbb{E}H_\nu = \mathbf{\Sigma}_{0,ii} - 1$ and $\nu! = 2$, and $\nu = \mathbf{e}_i + \mathbf{e}_j$ $(i < j)$, with $H_\nu = x_i x_j$, $\mathbb{E}H_\nu = \mathbf{\Sigma}_{0,ij}$ and $\nu! = 1$; the sum is $\frac{1}{2}\sum_i (\partial \mathbf{\Sigma}_{ii})^2 + \sum_{i<j} (\partial \mathbf{\Sigma}_{ij})^2 = \frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$. For $m = 4$ at fixed covariance: by Corollary 1.3(ii), $\partial_\theta \mathbb{E}[H_\nu] = \partial_\theta \boldsymbol{\kappa}_4[\nu]$ with ν read as an index multiset; the Frobenius norm over *ordered* tuples counts each multiset ν exactly $4!/\nu!$ times, so $\|\partial \boldsymbol{\kappa}_4\|_F^2 = \sum_\nu (4!/\nu!)(\partial \boldsymbol{\kappa}_4[\nu])^2$ and $\sum_\nu (\partial \boldsymbol{\kappa}_4[\nu])^2/\nu! = \|\partial \boldsymbol{\kappa}_4\|_F^2/24$. This is the paper's sentence "since the multiset ν collects $4!/\nu!$ ordered index tuples". A direction with $\partial_\theta \mathbb{E}[H_\nu] = 0$ for every ν has no information at any level; under Assumption 1.9 the law is determined by its moments (the moment problem is determinate when the moment generating function is finite near zero), so such a direction does not move P_0 at all.

1.11 The sample-complexity corollary and its three caveats

Corollary 1.20 (Local asymptotic minimax; the paper's Corollary 5)**.** *Fix $t > 0$ and a coordinate θ of lowest degree m with efficient information $I_t^{\mathrm{eff}}(\theta) > 0$. For M i.i.d. sequences observed at level t and any estimator sequence $\hat{\theta}_M$,*

$$\lim_{c\to\infty} \liminf_{M\to\infty} \sup_{|\theta'-\theta| \le c/\sqrt{M}} M \, \mathbb{E}_{\theta'}(\hat{\theta}_M - \theta')^2 \;\ge\; \frac{1}{I_t^{\mathrm{eff}}(\theta)} = \Theta(e^{2mt}),$$

and the maximum-likelihood estimator attains the bound.

This is the local asymptotic minimax theorem of Hájek and Le Cam (van der Vaart 1998, Theorem 8.11) for a family that is **differentiable in quadratic mean** at θ with non-singular information; at fixed $t > 0$ the family $\{P_t^\theta\}$ is smooth in θ (through Λ, under Assumption 1.9), which gives the differentiability, and the efficiency of maximum likelihood is the standard consequence of local asymptotic normality. The statement has the limit $c \to \infty$ outside: for a fixed c the local minimax risk is strictly smaller than $1/I$ (a bounded normal mean can be estimated better than the unrestricted one), and the paper's display, which fixes c, should be read with that limit understood. Three things the corollary does not say, all listed by the paper and all visible in its Section 4:

1. Nothing uniform in t. The regime is $M \to \infty$ at fixed t; local asymptotic normality needs $MI_t \gg 1$, i.e. $M \gg e^{2mt}$, so the limits in M and in t do not commute, and the $t \to \infty$ form of the constant is a second limit taken afterwards.

2. Nothing about absolute precision for a bounded coordinate. Once $1/\sqrt{MI_t}$ exceeds the range of the coordinate (the excess kurtosis of the paper's family lives on a bounded interval) the local statement is empty; this is the level at which the fitted shape in the paper's learning experiment collapses.

3. Nothing about an optimiser at a finite budget: EM at a validation-selected iteration count is not the maximum-likelihood estimator, and the paper reports where its iteration ladder binds.

The ratio statement of the paper, "a shape coordinate needs e^{4t} times more sequences than a covariance coordinate", is the ratio of two such bounds and inherits the same regime.

1.12 The crossover

For the Gaussian chain the expansion parameter is not α but $\alpha^2\lambda_{\max}(\mathbf{\Sigma}_0)$.

Proposition 1.21 (Three faces of t_*)**.** *Let $\mathbf{\Sigma}_0$ have largest eigenvalue $\lambda_{\max}$ and put $t_* = \frac{1}{2}\log\lambda_{\max}$.*

1. *The Neumann series $\mathbf{\Sigma}_t^{-1} = \sum_{n\geq 0}(-\alpha^2)^n(\mathbf{\Sigma}_0 - \mathbf{I})^n$ converges in operator norm if and only if $\alpha^2\|\mathbf{\Sigma}_0 - \mathbf{I}\| < 1$; when $\lambda_{\max} - 1 \geq 1 - \lambda_{\min}(\mathbf{\Sigma}_0)$ this is $e^{2t} > \lambda_{\max} - 1$.*
2. *$P_t/\gamma \in L^2(\gamma)$ if and only if $\lambda_{\max}(\mathbf{\Sigma}_t) < 2$, i.e. $e^{2t} > \lambda_{\max} - 1$; above that time the Hermite series Equation (1.17) converges in $L^2(\gamma)$, and below it it does not.*
3. *For the paper's chain, $\mathbf{\Sigma}_0 = (\rho^{|i-j|})_{ij}$ and $\lambda_{\max} \leq \sum_{k\in\mathbb{Z}}\rho^{|k|} = \frac{1+\rho}{1-\rho}$, so $t_* \leq \frac{1}{2}\log\frac{1+\rho}{1-\rho}$.*

Proof. (i) $\mathbf{\Sigma}_t = \mathbf{I} + \alpha^2(\mathbf{\Sigma}_0 - \mathbf{I})$ and the geometric series for $(\mathbf{I} + E)^{-1}$ converges iff $\|E\| < 1$; $\|\mathbf{\Sigma}_0 - \mathbf{I}\| = \max(\lambda_{\max} - 1, 1 - \lambda_{\min})$. (ii) $\int (P_t/\gamma)^2\gamma\,\mathrm{d}\mathbf{x} = \int P_t^2/\gamma\,\mathrm{d}\mathbf{x}$, and $P_t^2/\gamma \propto \exp(-\mathbf{x}^\top\mathbf{\Sigma}_t^{-1}\mathbf{x} + \frac{1}{2}|\mathbf{x}|^2) = \exp(-\frac{1}{2}\mathbf{x}^\top(2\mathbf{\Sigma}_t^{-1} - \mathbf{I})\mathbf{x})$, integrable iff $2\mathbf{\Sigma}_t^{-1} - \mathbf{I} \succ 0$ iff $\mathbf{\Sigma}_t \prec 2\mathbf{I}$ iff $\alpha^2\lambda_{\max} + \Delta_t < 2$ iff $\alpha^2(\lambda_{\max} - 1) < 1$. The $L^2(\gamma)$ norm of Equation (1.17) is $\sum_\nu \alpha^{2|\nu|}(\mathbb{E}H_\nu)^2/\nu!$, which is $\|P_t/\gamma\|^2_{L^2(\gamma)}$ by orthogonality, finite exactly then. (iii) $\mathbf{\Sigma}_0$ is a principal submatrix of the infinite Toeplitz matrix with symbol $g(\omega) = \sum_k \rho^{|k|}e^{ik\omega} = \frac{1-\rho^2}{1-2\rho\cos\omega+\rho^2}$, whose eigenvalues lie in $[\min g, \max g] = [\frac{1-\rho}{1+\rho}, \frac{1+\rho}{1-\rho}]$ (Szegő), and the eigenvalues of a principal submatrix interlace. □

At $\rho = 0.85$ the bound is $\frac{1}{2}\log(1.85/0.15) = \frac{1}{2}\log 12.33 = 1.26$; at $\rho = 0.5$ it is $\frac{1}{2}\log 3 = 0.55$; at $\rho = 0.95$ it is $\frac{1}{2}\log 39 = 1.83$. The exact $\lambda_{\max}$ at $L = 32$ is a little below the bound and is the value the paper's macro reports (Section 3, "Where the asymptote begins"; computed by `crossover_time` in `src/markov_score/theory.py`). Since $\lambda_{\max} > 2$ at the frozen correlation, $P_0/\gamma \notin L^2(\gamma)$ for the Gaussian chain itself, and P_t/γ enters $L^2(\gamma)$ only above $\frac{1}{2}\log(\lambda_{\max} - 1)$. For the Laplace chain P_t/γ is never in $L^2(\gamma)$: in one dimension a density with tails $\propto e^{-c|a|}$ convolved with a Gaussian has tails at least $\propto e^{-c|x|/\alpha}$ (restrict the convolution integral to $|a - x/\alpha| \leq 1$), so $P_t^2/\gamma \gtrsim e^{-2c|x|/\alpha + x^2/2}$ is not integrable; for the chain the site marginal has the same tail, and $\|P_t/\gamma\|_{L^2(\gamma)}$ dominates the marginal's norm by Jensen. This is the sentence "for the Laplace chain it never does, which is why the proof goes through Λ". Below t_* the long-wavelength modes of the covariance are still of order one and every rate is shallower than its asymptote; the paper uses this to explain the slopes of about -3 and -6 that earlier measurements on the model obtained over $t \leq 1.6$, below t_*, as pre-asymptotic values rather than as a failure of the law. For non-Gaussian laws the same scale is a heuristic that the measurements support.

1.13 Heavy tails

A density is **regularly varying** with index $\nu > 0$ if $\mathbb{P}(|\varepsilon| > u) = u^{-\nu}\ell(u)$ with ℓ slowly varying; the Student-t_ν density $\varphi(\varepsilon) \propto (1 + \varepsilon^2/((\nu - 2)q))^{-(\nu+1)/2}$, scaled to variance q, has $\varphi(\varepsilon) \sim c|\varepsilon|^{-\nu-1}$ and is the paper's example with $\nu = 5$. Its absolute moment of order k is finite if and only if $k < \nu$: t_5 has a fourth cumulant (excess kurtosis $6/(\nu - 4) = 6$) and an infinite sixth moment. Its moment generating function is infinite for every $s \neq 0$, so Assumption 1.9 fails and $\Lambda(s, 0) = +\infty$.

The $e^{-\nu t}$ term. The following heuristic, from the audit (Section 3.3) and the paper's "Heavy tails" paragraph, explains what replaces Equation (1.1). Take one site with unit variance. For $|x|$ large, $P_t(x) = \int P_0(a)\gamma_{\Delta_t}(x-\alpha a)\,\mathrm{d}a$ receives two contributions: from $a \approx x/\alpha$, where the prior's tail is slowly varying on the scale of the likelihood, giving $P_t(x) \approx P_0(x/\alpha)/\alpha \approx c\,\alpha^\nu |x|^{-\nu-1}$; and from $a \approx 0$, giving the Gaussian tail $\approx \gamma_{\Delta_t}(x)$. The first dominates once $|x| > K_t$ with $K_t^2 \approx 2\nu t$ (solve $\alpha^\nu |x|^{-\nu-1} = e^{-x^2/2}$ in the leading order). On that region the posterior of a concentrates near x/α, $\mathbb{E}[a \mid x] \approx x/\alpha - \Delta_t(\nu+1)/(\alpha x)$ (the prior's log-slope shifts the mean by $(\Delta_t/\alpha^2)\partial_a \log P_0(x/\alpha)$), so by Tweedie's identity $S(x) \approx -(\nu+1)/x$, while $S_{\mathrm{G}}(x) = -x$. The squared gap is $\approx x^2$ there, and its expectation over the region is

$$\int_{|x|>K_t} x^2\, c\,\alpha^\nu |x|^{-\nu-1}\,\mathrm{d}x = \frac{2c}{\nu-2}\,\alpha^\nu K_t^{2-\nu}\,:$$

a term of order $e^{-\nu t}$ up to a power of t. The probability of the region is $\approx \frac{2c}{\nu}\alpha^\nu K_t^{-\nu}$, also of order $e^{-\nu t}$, and it is carried by sequences with a site of size $|a| \approx |x|/\alpha \gtrsim K_t e^t$. In expectation the tail term leads once it exceeds $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$, which for $\nu < 8$ always happens eventually, late if $\|\boldsymbol{\kappa}_4\|_F^2$ is large; for $\nu > 8$ the Hermite term leads for ever, and finite eighth moments are the borderline of the conclusion, as remarked in Section 1.6. The paper states this as an observation supported by an untruncated scalar quadrature (E18) and not as a theorem.

Remark 1.22 (A discrepancy in the paper's wording). The paper writes that the tail term is "carried by sequences with $|\mathbf{a}| \gtrsim e^{2t}$, whose probability is itself of order $e^{-\nu t}$". For a tail of index ν the probability that a site exceeds e^{2t} is of order $e^{-2\nu t}$, not $e^{-\nu t}$; the events of probability $e^{-\nu t}$ that carry the term have $|a|$ of order e^t (times the slowly growing K_t), as derived above. The threshold e^{2t} is the boundary $|\mathbf{a}| > c/\alpha^2$ of the proof's bulk, which is a different object. One of the two numbers in the sentence should change.

What "never drawn" means. A sample of M sequences has ML sites, and the number of them beyond a threshold u is approximately Poisson with mean $ML\mathbb{P}(|a| > u)$. At $t = 3$ the carriers have $|a| \gtrsim K_t e^3 \approx \sqrt{30} \times 20 \approx 110$ in units of the site's standard deviation. For a variance-one t_5 variable, whose density is $\approx 47|a|^{-6}$ in standardised units for large $|a|$ (the constant is $\Gamma(3)5^{5/2}/(\sqrt{5\pi}\Gamma(5/2))$ times the scale factor $\sqrt{3/5}$ to the fifth power), $\mathbb{P}(|a| > 110) \approx 2 \times 9.5 \times (110/0.775)^{-5} \approx 3 \times 10^{-10}$, so a sample of 8192×32 sites contains, in expectation, about 10^{-4} carriers. A Monte Carlo estimate of the squared gap therefore reports the Hermite law $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$ at every level a finite sample can reach, and the $e^{-\nu t}$ term lives only in the expectation, which an untruncated quadrature can evaluate and a sample cannot. The paper's Figure (theory), panel (c), shows exactly this: the sampled Student-t_5 chain, on a domain wide enough to hold its sample, reaches the constant. What the earlier draft attributed to the moment condition on the narrow grid was the truncated prior in the reference (E18; the audit's Section 6.2).

1.14 The Edgeworth correction: adding the fourth-order term back

Equation (1.1) says what the twin lacks to leading order, and E16 (the script `experiments/e16_edgeworth_oracle.py`) adds it back exactly, with the true $\boldsymbol{\Sigma}_0$ and $\boldsymbol{\kappa}_4$, to test whether the gap is made of that term and nothing else.

Proposition 1.23 (First Edgeworth correction around the twin)**.** *Let $\mathbf{x}_t$ have covariance $\boldsymbol{\Sigma}_t$ and fourth-cumulant tensor $K = \alpha^4\boldsymbol{\kappa}_4$ (Corollary 1.3(i)), with vanishing odd cumulants. Then, formally, $P_t(\mathbf{x}) = P_t^{\mathrm{G}}(\mathbf{x})\big(1 + \epsilon(\mathbf{x})\big) + \dots$ with*

$$\epsilon(\mathbf{x}) = \frac{1}{24}\Big[K[y,y,y,y] - 6\,y^\top A y + 3\langle A, \boldsymbol{\Sigma}_t^{-1}\rangle\Big], \qquad y = \boldsymbol{\Sigma}_t^{-1}\mathbf{x}, \quad A_{ml} = \sum_{jk} K_{mjkl}(\boldsymbol{\Sigma}_t^{-1})_{jk}, \tag{1.18}$$

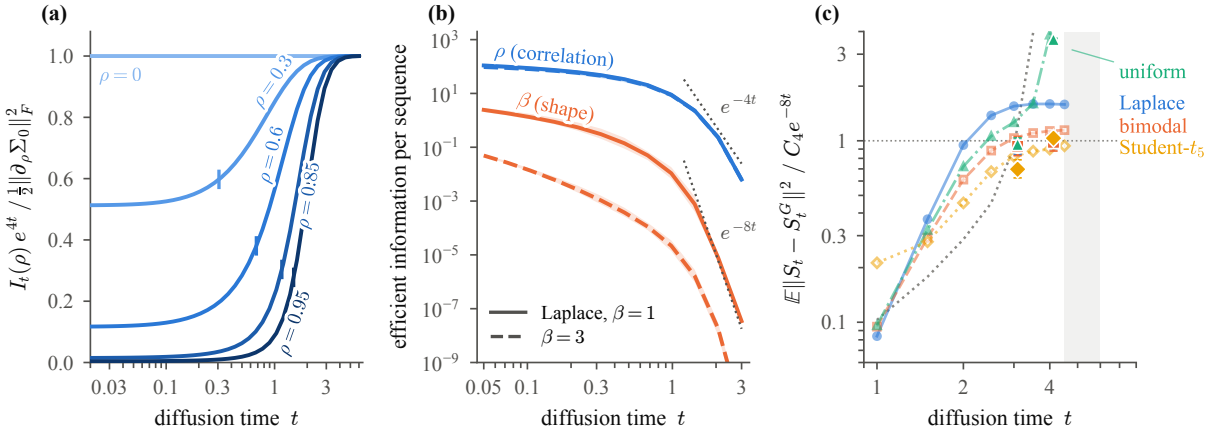


Figure 1.2: The paper's theory figure (Section 3), reproduced for reference. (a) Exact Fisher information of the Gaussian chain about ρ divided by its asymptote $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 e^{-4t}$, five correlations, ticks at $t_*(\rho)$ (Propositions 1.17 and 1.21). (b) Efficient information per sequence about ρ (degree two) and about the shape exponent β (degree four) of the generalised-Gaussian chain, from Fisher-identity scores, against the guides e^{-4t} and e^{-8t} (Section 1.9). (c) The closure gap $\mathbb{E}\|S - S_{\mathrm{G}}\|^2$ divided by $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$ for four innovation families (Proposition 1.12), with the large-sample constants and the double-precision floor.

and the corrected score is

$$S^{\mathrm{E4}} = S_{\mathrm{G}} + \frac{\nabla \epsilon}{1+\epsilon}, \qquad \nabla \epsilon = \tfrac{1}{6}\mathbf{\Sigma}_t^{-1} K[\cdot, y, y, y] - \tfrac{1}{2}\mathbf{\Sigma}_t^{-1} A y. \tag{1.19}$$

Proof. The characteristic function is $\varphi(\xi) = \exp(-\frac{1}{2}\xi^\top \mathbf{\Sigma}_t \xi + \frac{1}{24} K[\xi,\xi,\xi,\xi] + \dots) = \varphi^{\mathrm{G}}(\xi)(1 + \frac{1}{24} K[\xi,\xi,\xi,\xi] + \dots)$. Fourier inversion turns multiplication by $\xi_i\xi_j\xi_k\xi_l$ into $\partial_i\partial_j\partial_k\partial_l$ (the four factors of $-i$ give $(-i)^4 = 1$), so the correction to the density is $\frac{1}{24}\sum K_{ijkl}\partial_{ijkl} P_t^{\mathrm{G}}$. With $P_t^{\mathrm{G}} \propto \exp(-\frac{1}{2}\mathbf{x}^\top \mathbf{\Sigma}_t^{-1}\mathbf{x})$ and $y = \mathbf{\Sigma}_t^{-1}\mathbf{x}$, $P = \mathbf{\Sigma}_t^{-1}$, repeated differentiation gives $\partial_{ijkl} P_t^{\mathrm{G}} = [y_i y_j y_k y_l - (P_{ij} y_k y_l + 5 \text{ terms}) + (P_{ij}P_{kl} + 2 \text{ terms})]\, P_t^{\mathrm{G}}$, the Hermite tensor of $\mathcal{N}(0, \mathbf{\Sigma}_t)$. Contracting with the symmetric K gives the bracket of Equation (1.18). The score of $P_t^{\mathrm{G}}(1+\epsilon)$ is $S_{\mathrm{G}} + \nabla \log(1+\epsilon)$; and $\nabla_{\mathbf{x}} y = P$, so $\nabla K[y,y,y,y] = 4PK[\cdot,y,y,y]$ and $\nabla(y^\top A y) = 2PAy$ (A is symmetric because K is), giving Equation (1.19). □

At high noise $\mathbf{\Sigma}_t \to \mathbf{I}$, $y \to \mathbf{x}$ and $\epsilon \to \frac{\alpha^4}{24}\sum \boldsymbol{\kappa}_4[ijkl]{:}x_i x_j x_k x_l{:}$, which is exactly $\Lambda_4 - \Lambda_4^{\mathrm{G}}$ of Equation (1.16); the corrected score then differs from the true one at order α^6 instead of α^4, so the fraction of the squared gap it closes tends to one. At low noise $y = \mathbf{\Sigma}_t^{-1}\mathbf{x}$ is amplified by up to $1/\lambda_{\min}(\mathbf{\Sigma}_0)$, ϵ is a quartic of order one or larger, $1+\epsilon$ can approach or cross zero (the code clips it at 0.1), and the correction hurts: the series in α does not converge there. E16 measures the fraction $1 - \mathbb{E}\|S^{\mathrm{E4}} - S\|^2 / \mathbb{E}\|S_{\mathrm{G}} - S\|^2$ level by level on the Laplace, bimodal, uniform and Student-t_5 chains, with the "lin" form $S_{\mathrm{G}} + \nabla\epsilon$ as a control; the paper reports the fractions (Section 3, "The gap is the fourth-order term"; Appendix E) and the level below which the expansion diverges. Pooled over the twelve levels the corrected score is worse than the twin, because the pooled estimand is dominated by the lowest levels (**??**); the correction is a statement about what the gap is made of at each level, not a competitor on the schedule. On Student-t_5 the correction can close the Hermite part only, and the levels where it fails are the levels where the tail term of Section 1.13 leads.

1.15 Reading the theory figure

Figure 1.2 is the paper's Figure in Section 3. Panel (a) is Proposition 1.17: no Monte Carlo, a closed form on a fine grid of t (`part_gauss` of `experiments/e01_information_decay.py`); the curves

reach one, and the tick marks show that a strongly correlated chain approaches its asymptote later, as $\alpha^2\lambda_{\max}$ requires. Panel (b) is Propositions 1.15 and 1.16 measured on the chain: the per-sequence scores of the marginal likelihood are obtained by Fisher's identity through the tree-exact recursion (`per_chain_scores` in `src/markov_score/information.py`), the outer-product matrix is inverted for the efficient information, and the slopes above t_* are compared with -4 and -8; the constant of the shape rate is verified to high precision only in the scalar case (E21), because the outer-product estimate on the chain is a small residual of order-one quantities and reaches a quadrature floor. Panel (c) is Proposition 1.12: the gap is measured by tree-exact message passing against the LMMSE twin, divided by the predicted constant, on a domain wide enough for each family's sample (E18), with the bootstrap intervals from many sequences because the per-sequence gap is heavy-tailed (a degree-six polynomial in nearly Gaussian variables). The uniform family, with the smallest constant, reaches the double-precision floor of the posterior means first. All numbers are in the paper; none is repeated here.

1.16 Worked examples

Example 1.24 (The scalar Laplace law)**.** A unit-variance Laplace variable has excess kurtosis 3, so $\kappa_4 = 3$ and the constant of Equation (1.1) is $\kappa_4^2/6 = 1.5$. The leading term of the score gap is $\frac{3}{6}\mathrm{He}_3(x)e^{-4t} = \frac{1}{2}(x^3 - 3x)e^{-4t}$; at $x = 2$ and $t = 1$ it is $\frac{1}{2}\cdot 2\cdot e^{-4} = 0.0183$, against a twin score of -2. The expected squared gap at $t = 2$ is $1.5\,e^{-16} = 1.5 \times 1.13 \times 10^{-7} = 1.7 \times 10^{-7}$, and since the score energy at that level is close to one (**??**), the relative score error of the twin is about $\sqrt{1.7 \times 10^{-7}} = 4 \times 10^{-4}$.

Example 1.25 (The shape constant at the Gaussian member)**.** The generalised Gaussian $\varphi_\beta \propto e^{-|\varepsilon/s_\beta|^\beta}$ at unit variance has excess kurtosis $\kappa(\beta) = \Gamma(5/\beta)\Gamma(1/\beta)/\Gamma(3/\beta)^2 - 3$. Its derivative at the Gaussian member $\beta = 2$ can be done by hand with the digamma recursion $\psi(x+1) = \psi(x) + 1/x$: $\frac{\mathrm{d}}{\mathrm{d}\beta}\log(\kappa+3) = \beta^{-2}[-5\psi(5/\beta) - \psi(1/\beta) + 6\psi(3/\beta)]$, and at $\beta = 2$, with $\psi(5/2) = \psi(1/2) + 2 + \frac{2}{3}$ and $\psi(3/2) = \psi(1/2) + 2$, the bracket is $(-5 - 1 + 6)\psi(\frac{1}{2}) - \frac{40}{3} + 12 = -\frac{4}{3}$, so the derivative of the logarithm is $-\frac{1}{3}$ and $\partial_\beta\kappa|_{\beta=2} = 3 \times (-\frac{1}{3}) = -1$ exactly. In the scalar case Equation (1.3) therefore predicts $I_t(\beta) = \frac{1}{24}e^{-8t}(1 + O(e^{-2t}))$ at the Gaussian member: an information of one part in twenty-four times e^{-8t} per observation. The paper's E21 checks this constant by untruncated quadrature (Appendix E).

Example 1.26 (The two-site Gaussian chain, exactly)**.** For $L = 2$, $\mathbf{\Sigma}_0 = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ and $\partial_\rho\mathbf{\Sigma}_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so $\frac{1}{2}\|\partial_\rho\mathbf{\Sigma}_0\|_F^2 = 1$. With $\mathbf{\Sigma}_t = \left(\begin{smallmatrix} 1 & \alpha^2\rho \\ \alpha^2\rho & 1 \end{smallmatrix}\right)$, $\mathbf{\Sigma}_t^{-1}\partial_\rho\mathbf{\Sigma}_t = \frac{\alpha^2}{1-\alpha^4\rho^2}\left(\begin{smallmatrix} -\alpha^2\rho & 1 \\ 1 & -\alpha^2\rho \end{smallmatrix}\right)$, whose square has trace $2\alpha^4(1+\alpha^4\rho^2)/(1-\alpha^4\rho^2)^2$, so by Proposition 1.17

$$I_t(\rho) = \frac{\alpha^4(1+\alpha^4\rho^2)}{(1-\alpha^4\rho^2)^2} = e^{-4t}(1 + 3\alpha^4\rho^2 + O(\alpha^8)).$$

At $\rho = 0.85$ and $t = 1$: $\alpha^4 = e^{-4} = 0.01832$ and $\alpha^4\rho^2 = 0.01323$, so

$$I_1 = \frac{0.01832 \times 1.01323}{0.98677^2} = 0.01906,$$

four per cent above the asymptote. Here the correction is $O(\alpha^4)$ rather than $O(\alpha^2)$ because $\mathrm{tr}(\mathbf{\Sigma}_0 - \mathbf{I}) = 0$ and the first-order term of the Neumann series drops out of the trace; for the 32-site chain it does not, and the correction is $O(\alpha^2\lambda_{\max})$.

Example 1.27 (The fourth-cumulant tensor of a three-site chain)**.** Let $a_1 \sim \mathcal{N}(0,1)$, $a_2 = \rho a_1 + \varepsilon_2$, $a_3 = \rho a_2 + \varepsilon_3 = \rho^2 a_1 + \rho\varepsilon_2 + \varepsilon_3$, with i.i.d. innovations of variance $q = 1 - \rho^2$ and fourth cumulant $\kappa_4(\varepsilon) = 3q^2$ (Laplace). Joint cumulants are additive over the independent summands and a_1 is Gaussian, so $\boldsymbol{\kappa}_4[ijkl] = \kappa_4(\varepsilon)\sum_{m=2}^{\min(i,j,k,l)}\rho^{i+j+k+l-4m}$ (`chain_kappa4`

in `src/markov_score/theory.py`). The nonzero entries up to permutation are $\boldsymbol{\kappa}_4[2222] = 3q^2$, $\boldsymbol{\kappa}_4[2223] = 3q^2\rho$, $\boldsymbol{\kappa}_4[2233] = 3q^2\rho^2$, $\boldsymbol{\kappa}_4[2333] = 3q^2\rho^3$ and $\boldsymbol{\kappa}_4[3333] = 3q^2(\rho^4+1)$. Over ordered tuples the Frobenius norm is

$$\|\boldsymbol{\kappa}_4\|_F^2 = 9q^4[1 + 4\rho^2 + 6\rho^4 + 4\rho^6 + (1+\rho^4)^2],$$

the multiplicities being $1, 4, 6, 4, 1$. At $\rho = 0.85$: $q = 0.2775$, $q^4 = 0.00593$, $\rho^2 = 0.7225$, $\rho^4 = 0.5220$, $\rho^6 = 0.3772$; the bracket is $1 + 2.890 + 3.132 + 1.509 + 2.317 = 10.85$, so $\|\boldsymbol{\kappa}_4\|_F^2 = 9 \times 0.00593 \times 10.85 = 0.579$ and $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 = 0.0965$. The same recipe at $L = 32$ gives the constant the paper quotes for the Laplace chain (Section 3, "Verification with the exact score"), which grows linearly in L away from the first site.

Where it is used. Section 3 of the paper (Theorems 2–4, Assumption (E), Corollary 5, the crossover and heavy-tail paragraphs, the Edgeworth paragraph) and Appendix A (the proofs). Experiments: E01 (Gaussian closed form, shape information on the chain, closure gap), E16 (Edgeworth correction), E18 (closure constant and heavy tails on wide domains, untruncated scalar quadrature), E21 (information curve at $\rho = 0.5$ and the scalar shape constant). Code: `src/markov_score/theory.py` (`he`, `chain_covariance`, `chain_kappa4`, `gaussian_chain_fisher`, `score_gap_constant`, `shape_information_constant`, `correlation_information_constant`, `crossover_time`), `src/markov_score/information.py` (`per_chain_scores`, `outer_product_information`, `efficient_information`), `experiments/e01_information_decay.py`, `experiments/e16_edgeworth_oracle.py`, `experiments/e18_closure_wide.py`, `experiments/e21_information_curve.py`; the Lean targets for the constants are listed in `lean/README.md`. Audit: `docs/SCIENCE_AUDIT.md`, `reports/audit-2026-09-18.tex`, Section 3.

References. F. G. Mehler, 1866, "Ueber die Entwicklung einer Function von beliebig vielen Variabeln nach Laplaceschen Functionen höherer Ordnung", Journal für die reine und angewandte Mathematik 66, 161–176. S. Janson, 1997, *Gaussian Hilbert Spaces*, Cambridge University Press (Wick products and Wick's theorem, Chapter 3). P. McCullagh, 1987, *Tensor Methods in Statistics*, Chapman and Hall (joint cumulants, the Leonov–Shiryaev formula, Edgeworth expansions). A. W. van der Vaart, 1998, *Asymptotic Statistics*, Cambridge University Press (local asymptotic minimax theorem, Chapter 8). D. Guo, S. Shamai and S. Verdú, 2005, "Mutual information and minimum mean-square error in Gaussian channels", IEEE Transactions on Information Theory 51, 1261–1282. A. Katsevich and A. S. Bandeira, 2023, "Likelihood maximization and moment matching in low SNR Gaussian mixture models", Communications on Pure and Applied Mathematics 76, 788–842.